Gradient – Diagnostics

Equipment Room

Main Disconnect Panel CB3 TB1-15-16 CB2
Remote MRU
E-Off Switch
O_2 Monitor Control Unit — Meter, Ctrl Board, Alarm, Light
Customer-Supplied Outlet
Site E-Net
Facility Power Filters
Facility Utilities — Facility Coolant Supply, Facility Coolant Return

Operator Workspace

Temp/Hum Sensor
Server (Option)
Service Laptop
Optional Speaker
GOC — IRDC (Option) J5 J4 J3 J2 J1; USB-Serial Converter; Ethernet Switch 1 2; GOCAA J12 J14 J17 J18 J15 J7 J6
Host PC — Ethernet B, Lwr Graphic Card Port 1, RS232, USB, USB, USB, USB, Upr Graphic Card Port 1, Ethernet A, Line In, Line Out
GOC PDU — J10 J4 J6 J9 J2 J5 J1 J3
Global Modem (Option)
Key-board
Mouse
Primary Display
Pneumatic Patient Alert System

HEC

Mag Mon J5 J10 J8 J9 J7
HEC Power Box — Power Supply, Cryo, GC, PE, Blower
Signal Box J24 J11 J12
Grad HtXchr
Grad Res
Grad Pump
Blower HtXchgr
Grad/RF Coil Blower
Flow Meter
Filter
JR Power Cryo-Cooler Compressor JC — Water Return, Water Supply, Gas Return, Gas Supply
PGR Pump
PGR Res
PGR HtXchr

PGR

PDM J29 J25 L1/L2/L3 J27 J28 J24
VRE 2 (32-ch) J5 — Net 3, Net 2, Net Mgmt J1
VRE 1 J1 — Net Mgmt, Net 2, Net 3 J5
ETH 1 3 2 4 Term Server
Aux Board J4 — J6, J1 J2/J3 J9 J11 J10; XRFD LS1-A LS1-B LS2-A LS2-B LS3-A LS3-B LS4-A Term; PDU Top 1, PDU Top 1, PDU Front
J4 XGD Fan x3
CB1 J1 J4 J3 PGR PDU Host J6/J7/J8 J11/J12/J13 — Unused; Recon; RF/MNS J5 J2 PEN
CB2 J14 NB Amp J9 J10 J20 J5 J11 J7 J15 J19 J16 J8 J12 J6 J4 J3 J18 J1 J2
16 Port E-net Sw 3 2 1 4 5 6 14 15 12 11 13 7
J14 J15 J13 J12 J4 J3 J6 J102 J19 J18 J10 J9 J17 J2 J1
J3 J6

Cronus Chassis

XPS CB J13/J16 J12/J15 J11/J14 P1
XGA CB J13/J16 J12/J15 J11/J14
J10/J11 J6 J9 GMC J8 J7 J2 J1
Cronus Chassis Rear (Motherboard Connects)
Pwr XPS x3 J2 J10 J1 J11 J8 J9 J3 J4/J5 J4/J5
XGA x3 J8 J7 J16 J13 J15 J14 J5 J12 J6 J17 J18 J1/J2 J1/J2 J1/J2
Manifold Supply
SMA Coax 9-pin LEM Icoil x3 — Grad Pulse In, Grad Pulse Out
Term Block x3
Manifold Return

CAM

J8 NB J3 AIF
UPM1 — Det J1 J2 J4 J5 J3; Proc J3
UPM2 — J1 Det J2 J4 J5 J3; Proc J3
CAM — J2 ETH; SCP2 PMC J3; 10/100 Base; AGP2; SRF/TRF; STIF J4 J3 J18/J19; J23 I/O Panel J10; IRF3 J8; J14 J13 J12 J11
2 4 5 6 3 7 7 7

J8 J20 J204 J201 J200 J203 J202 J205 J22 J11 J21 J1 J31 J5 J16 J103 — Unused

Mag Bore

J6 J7 Body Coil J2 J3 J9
Bore-PAL L R
Left Arc Cover — Ambient Temp Snsr
Bore Temp Sensors
Front Mic
Airflow Sensors
X Y Z XRMB Gradient Temp Sensor
MAG-Grad Coil
MAG-Grad Coil
J101 J100
J104 Grad Coil Air
Patient Air

Magnet

J3 P403 Terminal Box
Pressure Transducer
Reed Switch
J2 RUO PreAmp J1
Coldhead Pwr Sleeve
Coldhead Supply Return
Airflow Sensor Module
J9/J10 J2 PAC
SRI4 — J23 J10 J34 J3 J20 J32 J24 J17 J1 J4 J5 J11/J12 J13/J14 J33 J21 J31 J2 J22
PreAmp & Combiner J11 J10 J9
J1 J5 J2 J6 J3 J7 BHYB J4
LCP J11 J3
RCP J11 J3
IRD J1
J11 J2 J9 J10 J12 TDM J1
J9 RRx1 J7 J1 J4/J5 J8 J2
J7 RRx2 (32-ch) J1 J4/J5 J2

Scan Room

MAG-PAC J4
PAC I/F
Squeeze Bulb
Speaker
Left E-Stop Button
Rt E-Stop Button

Patient Table

Coil Port P3
Coil Port P4
Touch N Go_R
Touch N Go_L
Table Hydraulics

Dock

J2 Mag
J1 Dock
J3 Left
ODU

Rear Pedestal

J14 J13 Dummy Load J12HYB
PED Antenna
Lights
RF Hub — Hub Backplane (bias, power, status/control); J102 J101 J104 J103
J1 J3 J2 J4 PDB J5 J6
MCDB
J9 J13 J12 J16 J17 J10 J14/J15 J7 J6 J1 J2 RFCB J8 J3 J4 — Debug Port
J1–J6 (ch 17-32)
RFSB (ch 1-16) J1–J6
J4 J6 J3 J1 J2 J5 TIF
Long Drive J1 ASM (MTR) J2
Load A
POS Enc
J3 EOT Sensor, HOME Sensor, EGRESS Sensor
LPCA — J4 J2 Port A J5 J1; Port P2 J4; Port P1 J5
Rear Mic
J1 PED I/O J2
Rear Pedestal Air
J10 J7 P2A P2B P1A P1B Reroute Box P3A P3B P4A P4B J8 J9
Load B

PEN

J14 J1 J13 J11 J12 J10 F/O Wvgde
J18 J7 J23 J22 J19 J18 J15 J12 NB Exciter J19 J20 J21 J29 J32 J30 J15 J14 J17 J1 J17
J1 Power Distribution J4 J2 J3
J18 J19 J7/J8 J1/J2 J5/J6 J3/J4 CFB J17
J12 J9 J37/J38 J2 J43 J11 J39 J54 J41 J1 J5 J40 J10 PHPS J42 J35 J36 J44 J8 J39 J54 J41 J40 J42 J35 J36
J14 J8 J19 J17 J11 J20 J12 Driver Module J1 J13 J2 J7 J6 J23 J25 J21
J6 J51 J52 J46 J48 J45 J50 J47 J5 J1 SRPS J3 J51 J52 J46 J48 J45 J50 J47 J29 J32 J30 J34
J94 J96 J92 DD Filter J74 J76 J72
J4 J9 J5 J6 J16 J3 J4 J9 J5 J6 J16 J11 J31 J33 F11 F10 F9 F8 F7 F6 F5

Temp/Hum Sensor
Scan Room Blower Box
Patient Comfort
Hub

Mag I/O

J1 J3 J6 J4 J2 J5 J14 J15 J12 J11 J13 J8 J7 J9 J10 J17 J16

FO I/F

J28 J20 J21

Path Info

When diagnostics are initiated by the FE,
1. Status/control path originates in the Host, and passes to the Ethernet Switch, then
2. To the SCP. The SCP translates the command and passes it to either the Ethernet Switch or the SRF/TRF board (depending on which diagnostic is requested).
2,3. When a Static Test or Fidelity Diagnostic is requested, the SCP passes the translated command to Cronus Chassis GMC board via the Ethernet Switch, and the GMC board executes the command.
4. When a Gradient Hammer Test is requested, the SCP passes the translated command to the SRF/TRF board, which executes the command.

Real-time diagnostic execution involves the following path:
5. From the SRF/TRF to the IRF3, then
6. To the Cronus Chassis GMC board, then,
7. To the XGA Control Board, via the Cronus backplane, and to the XGAs
So if there are multiple failures, the failure in the later steps will not come to light until the failure in the earlier step has been addressed.

Data Path – General

The Host PC is connected to the Gradient chain through the Ethernet Switch. The Gradient information, in turn, passes through the Ethernet Switch to the CAM Chassis via the AGP Board. From the AGP board, information is passed to the SRF/TRF board via the CAM backplane. The SRF/TRF board operates as the brains of the process, passing the signal on to the other two boards via the CAM backplane.

Once fully processed the signal is sent to the Cronus Chassis in the XGD before the Gradient Pulse is generated.

Door Switch
Coolant Removal Kit

Key

TBD = To Be Determined (Future Enhancement)

Cabinet
Chassis/ Sub-system
Part
Data Cables
Coolant
Power
Pulse
Data Paths

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